

Chapter 1

$\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$

The set $\mathbb{R}^n$

- $\mathbb{R}$ is the set of all real numbers.
- Let n be a positive integer.
- We can write $(x_1, x_2, \dots, x_n)$ of real numbers, which is an ordered n -tuple of real numbers.
- $\mathbb{R}^n$ is the set of all ordered n -tuples of real numbers.
- An element $x \in \mathbb{R}^n$ can be written in

$$\text{row } x = (x_1, x_2, x_3, \dots, x_n) \quad \text{or in column } x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{or any other form.}$$

As long as the order of listed n real numbers are seen, there would not be a problem.

Addition and Scalar multiplication

Addition

1. If $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and $y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$, we can add the two to obtain an element in $\mathbb{R}^n$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \\ \vdots \\ x_n + y_n \end{pmatrix},$$

which is denoted by $x + y \in \mathbb{R}^n$.

Scalar multiplication

1. In this course, the scalar is a synonym of real number.
2. If λ is a real number and $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, then we can scale x to obtain λx :

$$\lambda \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \\ \vdots \\ \lambda x_n \end{pmatrix} \in \mathbb{R}^n.$$

The set $\mathbb{R}^{m \times n}$

- Let $\mathbb{R}$ be again the set of all real numbers.
- Let m and n be two positive integers.
- The set of all $(A_{ij})_{i=1, j=1}^{i=m, j=n}$ of real numbers indexed by $(i, j) \in \{1, 2, \dots, m\} \times \{1, 2, \dots, n\}$ is denoted by $\mathbb{R}^{m \times n}$.
- The notation for an element $A \in \mathbb{R}^{m \times n}$ in this time is more specific. We distinguish two indices $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$. We list mn real numbers in a box so that i is a row index, and j is a column index:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix}.$$

- We call an element of $\mathbb{R}^{m \times n}$ an $(m \times n)$ matrix, which reads as “ m by n matrix”.
- Why ..?

Addition and Scalar multiplication

Addition

1. If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times n}$, i.e., if the matrix A and B are in same shape, we can add the two to obtain an element in $\mathbb{R}^{m \times n}$:

$$\begin{aligned} & \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix} + \begin{pmatrix} B_{11} & B_{12} & B_{13} & \cdots & B_{1n} \\ B_{21} & B_{22} & B_{23} & \cdots & B_{2n} \\ B_{31} & B_{32} & B_{33} & \cdots & B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ B_{m1} & B_{m2} & B_{m3} & \cdots & B_{mn} \end{pmatrix} \\ &= \begin{pmatrix} A_{11} + B_{11} & A_{12} + B_{12} & A_{13} + B_{13} & \cdots & A_{1n} + B_{1n} \\ A_{21} + B_{21} & A_{22} + B_{22} & A_{23} + B_{23} & \cdots & A_{2n} + B_{2n} \\ A_{31} + B_{31} & A_{32} + B_{32} & A_{33} + B_{33} & \cdots & A_{3n} + B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} + B_{m1} & A_{m2} + B_{m2} & A_{m3} + B_{m3} & \cdots & A_{mn} + B_{mn} \end{pmatrix}, \end{aligned}$$

which is denoted by $A + B \in \mathbb{R}^{m \times n}$.

Scalar multiplication

1. If λ is a real number and $A \in \mathbb{R}^{m \times n}$, then we can scale A to obtain λA :

$$\lambda \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix} = \begin{pmatrix} \lambda A_{11} & \lambda A_{12} & \lambda A_{13} & \cdots & \lambda A_{1n} \\ \lambda A_{21} & \lambda A_{22} & \lambda A_{23} & \cdots & \lambda A_{2n} \\ \lambda A_{31} & \lambda A_{32} & \lambda A_{33} & \cdots & \lambda A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \lambda A_{m1} & \lambda A_{m2} & \lambda A_{m3} & \cdots & \lambda A_{mn} \end{pmatrix}.$$

In summary,

1. We have the set $\mathbb{R}^n$, equipped with the addition and the scalar multiplication, that we denote by $(\mathbb{R}^n, +, s)$.

$$\begin{aligned} + : \mathbb{R}^n \times \mathbb{R}^n &\rightarrow \mathbb{R}^n, \\ s : \mathbb{R} \times \mathbb{R}^n &\rightarrow \mathbb{R}^n. \end{aligned}$$

2. We have the set $\mathbb{R}^{m \times n}$, equipped with the addition and the scalar multiplication, that we denote by $(\mathbb{R}^{m \times n}, +, s)$.

$$\begin{aligned} + : \mathbb{R}^n \times \mathbb{R}^n &\rightarrow \mathbb{R}^n, \\ s : \mathbb{R} \times \mathbb{R}^n &\rightarrow \mathbb{R}^n. \end{aligned}$$

1. That we work with $(\mathbb{R}^n, +, s)$, i.e., that $\mathbb{R}^n$ are equipped with the addition and the scalar multiplication may, more importantly, mean that we do not perform other operations.

These are illegal expressions:

- (a) $\mathbb{R}^n + \mathbb{R}$, $n \geq 2$:
- (b) $\mathbb{R}^n + \mathbb{R}^m$, $n \neq m$.
- (c) product in general. $\times : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, though in some dimensions we can define meaningful product.
- (d) comparison in general.

2. Likewise, for the case of $\mathbb{R}^{m \times n}$, we do not perform other operations, unless otherwise defined later of this course.

- We have so far

$$(\mathbb{R}^n, +, s) \quad \text{and} \quad (\mathbb{R}^{m \times n}, +, s),$$

respectively a set equipped with the addition and scalar multiplication.

- Expression that makes use of the addition and scalar multiplication, and use only of them such as

$$\lambda_1 x + \lambda_2 y + \lambda_3 z + \lambda_4 w, \quad x, y, z, w \in \mathbb{R}^n$$

comes as an important expression to us.

- The expression is called a “Linear Combination” of x, y, z, w ; Linear, meaning that the scalar multiplication $x \mapsto \lambda x$ is to make something proportional, and Combination, meaning here we are adding things.

The final comment on $(\mathbb{R}^n, +, s)$ and $(\mathbb{R}^{m \times n}, +, s)$:

As stated earlier, in some dimensions we may additionally define product, for example in $\mathbb{R}^2$ we can define

$$(x_1, x_2) \times (y_1, y_2) = (v_1, v_2), \quad \text{where} \quad v_1 + iv_2 = (x_1 + ix_2)(y_1 + iy_2).$$

However, all of such operations are treated in this course to be exceptional and come as addendum, emphasizing that there are only (i) the addition and (ii) the scalar multiplication we are allowed to operate so far.

Chapter 2

Matrix Matrix multiplication

- Emphasizing that there are no other general product operations, now we define the unique product operation between matrices.
- For given matrix $A \in \mathbb{R}^{\ell \times m}$ and matrix $B \in \mathbb{R}^{m \times n}$, we define the matrix $C \in \mathbb{R}^{\ell \times n}$ to be the product denoted by AB so that

$$C_{ij} = \sum_{\alpha=1}^m A_{i\alpha} B_{\alpha j} \quad \text{for } i = 1, 2, \dots, \ell \text{ and } j = 1, 2, \dots, n.$$

- Importantly, the product is defined only for the case where the second component of shape of A and the first component of shape of B are the same, and the product is not defined for all remaining cases.
- Why .. ?

The product is useful in many places.

Perfect for the chain rule in multivariable calculus

Suppose that $X = \mathbb{R}^n$, $Y = \mathbb{R}^m$, and $Z = \mathbb{R}^\ell$, and

$$f : X \rightarrow Y, \quad g : Y \rightarrow Z.$$

Multivariable Calculus:

1. Also assume f and g are many times differentiable functions.
2. For a given point $\bar{x}$, we collect numbers

$$B_{\alpha j} = \frac{\partial f^\alpha}{\partial x^j}(\bar{x}) = \lim_{h \rightarrow 0} \frac{f^\alpha(\bar{x} + h e_j) - f^\alpha(\bar{x})}{h}.$$

3. Similarly, for a given point $\bar{y}$, we collect numbers

$$A_{i\alpha} = \frac{\partial g^i}{\partial y^\alpha}(\bar{y}) = \lim_{h \rightarrow 0} \frac{g^i(\bar{y} + h e_\alpha) - g^i(\bar{y})}{h}.$$

4. Now we consider the composition

$$g \circ f : X \rightarrow Z, \quad g \circ f(x) = g(f(x))$$

and collect numbers

$$C_{ij} = \frac{\partial (g \circ f)^i}{\partial x^j}(\bar{x}) = \lim_{h \rightarrow 0} \frac{(g \circ f)^i(\bar{x} + h e_j) - (g \circ f)^i(\bar{x})}{h}.$$

5. Then C turns out to equal to AB , for $\bar{y} = f(\bar{x})$.

Writing a system of linear equations

If we are given

$$\begin{cases} 3x - 7y + 4z = -2, \\ 9x - 2y - 6z = 0, \\ -5x + 3y - 11z = -8 \end{cases}$$

It is nice that we can write also

$$\begin{pmatrix} 3 & -7 & 4 \\ 9 & -2 & -6 \\ -5 & 3 & -11 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ -8 \end{pmatrix}$$

Using $\ell = 3$, $m = 3$, and $n = 1$ in the left-hand-side.

The question of why..?

1. Why does it have to be in that way?
2. Why not we define for $A, B \in \mathbb{R}^{m \times n}$ a product with the same shape in $\mathbb{R}^{m \times n}$?

$$\begin{pmatrix} A_{11}B_{11} & A_{12}B_{12} & A_{13}B_{13} & \cdots & A_{1n}B_{1n} \\ A_{21}B_{21} & A_{22}B_{22} & A_{23}B_{23} & \cdots & A_{2n}B_{2n} \\ A_{31}B_{31} & A_{32}B_{32} & A_{33}B_{33} & \cdots & A_{3n}B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1}B_{m1} & A_{m2}B_{m2} & A_{m3}B_{m3} & \cdots & A_{mn}B_{mn} \end{pmatrix}$$

- (a) Actually you can. In most of computer language, this entry-wise product in the same shape is provided by the broadcasting.
3. All why-questions come to the following one,
Would other people be interested in the new definition ?

A few important observations

1. Important: You don't mess up with the order of multiplication here.
2. Unlike multiplication we know for two real numbers, for the Matrix-Matrix multiplication for $A \in \mathbb{R}^{\ell \times m}$ and $B \in \mathbb{R}^{m \times n}$, the product BA is not even defined in general. This is simply because in general $n \neq \ell$.
3. In the special case of that $\ell = m = n$, BA is defined. However even in such a case

$$AB \neq BA \quad \text{in general.}$$

Summary

1. We have $(\mathbb{R}^{m \times n}, +, s)$ for m and n positive integers.
2. Linear combinations of members of $\mathbb{R}^{m \times n}$ are expressions using $+$ and s such as

$$\lambda_1 A_1 + \lambda_2 A_2 + \cdots \lambda_j A_j \in \mathbb{R}^{m \times n}.$$

3. We have a definition of multiplication for $A \in \mathbb{R}^{\ell \times m}$ and matrix $B \in \mathbb{R}^{m \times n}$ resulting in $AB \in \mathbb{R}^{\ell \times n}$.

Knowledge so far will be applied to mathematics, science, and engineering, and will be extremely important and powerful.

Chapter 3

Linear Independence and Basis

Let us consider elements in $\mathbb{R}^2$,

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad z = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

It is easy to notice that if we are with x and y in our hands, z can be reproduced by

$$z = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

a linear combination of x and y .

Linear Combinations and Span

For a given finite number of elements $a_1, a_2, \dots, a_k \in \mathbb{R}^n$, the set of all linear combinations of them is denoted by

$$\text{span} \langle a_1, a_2, \dots, a_k \rangle = \{x \in \mathbb{R}^n \mid x \text{ is a linear combination of } a_1, a_2, \dots, a_k\}.$$

Example

Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \quad \cdots \quad e_n = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Then, $e_1, e_2, \dots, e_n$ span all elements in $\mathbb{R}^n$, i.e., $\text{span} \langle e_1, e_2, \dots, e_n \rangle = \mathbb{R}^n$.

Linear Independence

As far as reproducibility is concerned, we notice that we keep only two elements

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

and every element in $\mathbb{R}^2$ can be reproduced by linear combination of x and y .

This necessitates the notion of linear independence of a few elements in $\mathbb{R}^n$.

Example 1 z above is linearly dependent with respect to x and y .

Example 2 On the other hand, in $\mathbb{R}^3$,

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \quad z = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

Then, x, y, z are linearly independent. By no chance, z is a linear combination of x and y .

Definition 1. We say $a_1, a_2, \dots, a_k \in \mathbb{R}^n$ are linearly independent if

$$\lambda_1 a_1 + \lambda_2 a_2 + \dots + \lambda_k a_k = 0 \implies \lambda_1 = \lambda_2 = \dots = \lambda_k = 0.$$

$a_1, a_2, \dots, a_k \in \mathbb{R}^n$ are said to be linearly dependent if they are not linearly independent.

Basis

Definition 2. We say $a_1, a_2, \dots, a_k$ of $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$ if

1. $a_1, a_2, \dots, a_k$ span $\mathbb{R}^n$,
2. $a_1, a_2, \dots, a_k$ are linearly independent.

Examples

$e_1, e_2, \dots, e_n$ form a basis.

We will prove in later section the following seemingly natural facts:

1. Elements in $\mathbb{R}^n$ fewer than n cannot span all of $\mathbb{R}^n$.
2. Elements in $\mathbb{R}^n$ exceeding n must linearly dependent.
3. n elements in $\mathbb{R}^n$ are linearly independent if and only if they do span all of $\mathbb{R}^n$.

Chapter 4

Power and Importance of $\mathbb{R}^{m \times n}$

We saw that $(\mathbb{R}^{m \times n}, +, s)$ is so simply defined. The power of $\mathbb{R}^{m \times n}$ comes from that, (i) there are so many instances in mathematics, science, engineering, and etc., where entities of interests are such simple that we can add and scale them and that (ii) we will learn nice tools for $\mathbb{R}^{m \times n}$.

Example 1 In fact, this is an example of $\mathbb{C}^{m \times n}$, with $\mathbb{C}$ of complex numbers. Wolfgang Pauli, a physicist, introduced *Pauli matrices* in developing theory of quantum mechanics in early 20c.

We consider a multiplication table for a 4 elements set $\{1, \sigma_x, \sigma_y, \sigma_z\}$:

Table 4.1: A multiplicative table

	1	σ_x	σ_y	σ_z
1	1	σ_x	σ_y	σ_z
σ_x	σ_x	-1	σ_z	$-\sigma_y$
σ_y	σ_y	$-\sigma_z$	-1	σ_x
σ_z	σ_z	σ_y	$-\sigma_x$	-1

Note that $\sigma_x \sigma_y \neq \sigma_y \sigma_x$, and this set cannot be represented by elements in $\mathbb{R}$.

One of the way we achieve the table:

We consider

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \right\}$$

for $1, \sigma_x, \sigma_y, \sigma_z$ from the left.

The set $\text{span} \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \right\rangle$ contains all the elements that can be generated.

note: The role played by $\mathbb{R}$, the scalar factor, can be taken by the mathematical object of *field*. But we will stick to $\mathbb{R}$ in this course.

Example 2 Colors (Light)

Although there are so many colors such as yellow, green, blue, white, black, $\dots$, it turns out that the only three of them, red, green, and blue are independent. All others are linear combination of the three.

We choose $[0, 255]^3 \subset \mathbb{R}^3$ represents the colors, with

1. $(255, 0, 0) \simeq$ red,
2. $(0, 255, 0) \simeq$ green,
3. $(0, 0, 255) \simeq$ blue,
4. $(0, 0, 0) \simeq$ black,
5. $(255, 255, 255) \simeq$ white.

The value ranged in $[0, 255]$ represents the intensity.

Example 3 Bacteria with manipulatable genes.

In Bio-Engineering, certain bacteria is called a *vector*. This reflects that a certain kind of bacteria may or may not contain a few genes of our interests, and can be manipulated in the laboratory in the way of addition.

By a reproduction, a bacteria with gene A and a bacteria with gene B can result in a bacteria with both genes A and B . The number of such genes that can be manipulated in lab. and bacteria can be represented by $\mathbb{R}^n$.

Example 4 Sauce

Sauce can be perfectly represented by $\mathbb{R}^n$. They can be added and scaled:

We linear combine the sugar, salt, vinegar, soy sauce, water, what you want, in grams.

So a recipe you have to remember for a certain sauce is an element in $\mathbb{R}^n$, and if you want to make sauce a lot, you scale the element.

1. We saw many examples of *entities* that can be added and scaled, and sometimes we employ multiplication.
2. They are well-represented by $\mathbb{R}^{m \times n}$ or $\mathbb{R}^n$.
3. In all of examples, we see that there are two different roles:

Representation vs. Essential Entities

4. The theory has been developed in Linear Algebra so that the two roles are explicitly detached from each other. The latter, of essential entities, is defined in mathematics as *vector space* and *algebra*.
5. In our course, we use the word *vector* to denote the essential entity, while we simply use the word element of $\mathbb{R}^n$, or the matrix in $\mathbb{R}^{m \times n}$ for the representational use.
6. The connection from the set of essential entities to for instance $\mathbb{R}^n$, is done by fixing the basis. Let us be clear on this below.

- The definition of vector space X over the scalar $\mathbb{R}$ can be found in math textbooks and webpages. For completeness, we also contained the mathematical definition in the last page of this chapter.
- If you are not familiar with mathematical definitions, consider a vector space X as a set of symbols that can be added and scaled.
- k -dimensional vector space for k a positive integer.

Definition 3. A vector space X that admits a basis consists of k elements is called a k -dimensional vector space.

Example The space of colors admits a basis of three elements. Hence the space is 3-dimensional vector space.

- Connection from the *essential entities* to *representing elements* is done once we fix what are the basis elements in X .

Example The space of sauces consists of $a = \text{sugar}$, $b = \text{salt}$, $c = \text{vinegar}$, and $d = \text{water}$.

Once we fix a , b , c , and d to be the basis, we will understand an element

$$\begin{pmatrix} 5 \\ 5 \\ 3 \\ 100 \end{pmatrix} \in \mathbb{R}^4$$

to represent a sauce consisting of 5-sugar, 5-salt, 3-vinegar, and 100 water, in grams.

We may want to take the diluted ingredients $a + 100d$, $b + 100d$, $c + 100d$, and d to be the basis. Then

$$\begin{pmatrix} 5 \\ 5 \\ 3 \\ 100 \end{pmatrix} \in \mathbb{R}^4$$

to represent a sauce consisting of 5-sugar, 5-salt, 3-vinegar, and 1400 water, in grams.

- You can see the detached roles of *representation* and *essential entities*.
- And how the basis choice connects the two.

A vector space X over the scalar $\mathbb{R}$ is mathematically defined as follows.

1. X is a nonempty set.
2. $+: X \times X \rightarrow X$ is well-defined with following properties.
 - (a) For any $x, y, z \in X$, $x + (y + z) = (x + y) + z$.
 - (b) For any $x, y \in X$, $x + y = y + x$.
 - (c) $\exists 0 \in X$ such that for any $x \in X$ $x + 0 = x$.
 - (d) For any $x \in X$, $\exists(-x) \in X$ such that $x + (-x) = 0$.
3. $s: \mathbb{R} \times X \rightarrow X$ is well-defined with following properties.
 - (a) For any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $x \in X$, $\lambda_1(\lambda_2 x) = (\lambda_1 \lambda_2)x$
 - (b) For any $x \in X$, $1x = x$.
4. $+$ and s work in compatible ways satisfying the following.
 - (a) For any $\lambda \in \mathbb{R}$ and $x, y \in X$, $\lambda(x + y) = \lambda x + \lambda y$.
 - (b) For any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $x \in X$, $(\lambda_1 + \lambda_2)x = \lambda_1 x + \lambda_2 x$.

Summary

1. For an m -dimensional vector space $(X, +, s, \{b_1, b_2, \dots, b_m\})$ with m basis vectors $b_1, b_2, \dots, b_m$, we consider the representation of its element for example

$$\begin{pmatrix} 3 \\ -5 \\ 0 \\ \vdots \\ 0.8 \end{pmatrix} \in \mathbb{R}^{m \times 1}$$

to represent $3b_1 - 5b_2 + 0b_3 + \dots + 0.8b_m$.

2. Be careful when X itself is $\mathbb{R}^m$, which is the very possible example. One way we will follow in this course is that, for the representational use, we make use of

$$\mathbb{R}^{m \times 1} \quad \text{of one column matrices.}$$

and try best to distinguish the essential entity X and its representation.

Chapter 5

Transformation of Solid shapes

We consider the three dimensional space we live.

1. We consider the vector space X and algebra L of essential entities below.
2. Their respective representation will be $\mathbb{R}^{3 \times 1}$ and $\mathbb{R}^{3 \times 3}$.

We do the following for solid shapes:

The vector space X

1. We let the set

$$X = \{\text{Locations}\} \quad \text{with respect to a fixed origin.}$$

2. Scalar multiplication: Location with respect to an origin can be scaled by a real number.
3. Addition: take the parallelogram law.

Basis selection

1. We take the length 1 orthogonal locations e_1 , e_2 , and e_3 be the basis of X .
2. Then every location is a linear combination of those three locations.

Representation in $\mathbb{R}^{3 \times 1}$ columns

1. The coordinates of a location is then represented in $\mathbb{R}^{3 \times 1}$.

A solid shape is then a subset occupying a region of locations.

$$G \subset X$$

Linear Transformations

- A transformation is then a mapping

$$T : X \rightarrow X.$$

- We say $T : X \rightarrow X$ is linear if

$$\begin{aligned} (i) \quad & T(\lambda x) = \lambda T(x) \\ (ii) \quad & T(x + y) = T(x) + T(y). \end{aligned}$$

i.e., $T(\lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_k x_k) = \lambda_1 T(x_1) + \lambda_2 T(x_2) + \cdots + \lambda_k T(x_k)$.

- We then collect

$$L = \{T : X \rightarrow X \mid T \text{ is linear}\}.$$

Basis selection in L

1. There exists a linear mapping $\hat{T}^{1,1}$ such that

$$\hat{T}^{1,1}(e_1) = e_1, \quad \hat{T}^{1,1}(e_\beta) = 0 \quad \beta = 2, 3.$$

2. Similarly there exists a linear mapping $\hat{T}^{i,\alpha}$ such that

$$\hat{T}^{i,\alpha}(e_\alpha) = e_i, \quad \hat{T}^{i,\alpha}(e_\beta) = 0 \quad \beta \neq \alpha.$$

3. The 9 elements for $i = 1, 2, 3$ and $\alpha = 1, 2, 3$ are the basis of L

Representation in $\mathbb{R}^{3 \times 3}$

1. With such 9 basis, a linear transformation is represented in $\mathbb{R}^{3 \times 3}$.
2. For example

$$\hat{T}^{1,1} \quad \text{is represented by} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since we have fixed basis and representations, from now on, we work with numbers and matrices.

i.e., we have $x \in \mathbb{R}^{3 \times 1}$ and $A \in \mathbb{R}^{3 \times 3}$, by which, the new location is represented by

$$y = Ax \in \mathbb{R}^{3 \times 1}.$$

Here, we list a few elementary matrices, that does:

Dilation, Rotation, Projection, Reflection, and Shearing.

Identity Matrix

1. We consider the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

2. It is a special element in $\mathbb{R}^{3 \times 3}$, and denoted by I .

3. Note that for any $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$,

$$y = Ax = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

- Diagonal entries are numbers in diagonal of the square.
- Off diagonal entries are entries not in diagonal.
- All of diagonal entries of I are 1
- All of off diagonal entries of I are 0.

4. Be careful that I is not the one below !

$$I \neq \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Diagonal Matrices

1. Let $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$. We consider a matrix

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

2. First, let us consider the special cases $\lambda_1 > 0, \lambda_2 > 0, \lambda_3 > 0$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

3. $\lambda_i = 0$: One component is shrinken to 0.
4. $\lambda_i < 0$: Dilate with sign flip.
5. More matrices for the dilations below will be introduced later.
- (a) Principal axes are not coordinate axes.
- (b) Dilation with directions not orthogonal to each other.

Elementary Rotation Matrices

1. Let $\theta \in \mathbb{R}$. We consider a matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

- (a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$
- (b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$
- (c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$
- (d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

2. We consider a matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

3. We consider a matrix

$$\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

Elementary Orthogonal Projection Matrices

1. We consider a matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

2. We will introduce more orthogonal projection matrices onto the projecting plane that is not aligned with coordinate.
3. There are non-orthogonal projections, but we will not use them much. We will introduce them later.

Elementary Reflection Matrices

1. We consider a matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

2. We will introduce more reflection matrices whose symmetry hyperplane is not aligned with coordinate.

Upper Triangular matrices: Elementary Shearing

1. We consider a matrix

$$\begin{pmatrix} \lambda_1 & \alpha & \beta \\ 0 & \lambda_2 & \gamma \\ 0 & 0 & \lambda_3 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

2. This kind of matrices are called upper triangular matrices.

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ $Ax =$

We can consider a few consecutive applications of linear transformation,

$$T = A_k A_{k-1} A_{k-2} \cdots A_2 A_1$$

of 3×3 matrices.

1. We examined finitely many 3×3 elementary matrices that do:
 dilation, rotation, projection, reflection, shearing
 but only with respect to three axes or to xy , yz , and zx planes.
 2. Now, we learn the technique to incorporate those 5 deformations with axes not along the coordinate axes.
 3. To be concrete, let us consider a reflection with respect to a plane passing the origin that is not xy , yz , or zx plane.
-
1. Having seen the plane, we became to think: if I knew the reflection was what I was going to do, I should have selected three locations of the basis adapted to that plane.
 2. Now, let us pose another observer who happened to choose the basis as I wished, say the location f_1 , f_2 , and f_3 .
 3. He of course use $\mathbb{R}^{n+1}$ to represent the location, but in his language, this is a linear combination of his basis.

$$\begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \text{ refers to the location } 1f_1 - 3f_2 + 4f_3.$$

The only knowledge we need to translate his language

1. What if I want to know the location for

$$\begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

in my language ?

2. Let us make analogy, saying his language is english, and our language is korean.
 We would need

English-to-Korean Dictionary or Korean-to-English Dictionary

3. If the location f_1 , f_2 , and f_3 are known in my language, for example

$$f_1 = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}, \quad f_2 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}, \quad f_3 = \begin{pmatrix} -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \end{pmatrix}$$

This is going to be the English-to-Korean Dictionary. Let us list

$$F = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix}$$

that is my English-to-Korean Dictionary.

4. If so, another making use of Matrix-Matrix multiplication:

$$\text{the location in my language} = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} = 1 \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix} - 3 \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix} + 4 \begin{pmatrix} -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \end{pmatrix}.$$

5. In summary, if I know the English-to-Korean Dictionary $\begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix}$ and I know the location in English $\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}$ then the location in Korean $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is in relation

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}.$$

Now, it is time for Korean-to-English Dictionary

1. Now, I want to write things in English, for given a location in my language, I want to know how it is represented in his language.
2. That is of course,

$$\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} = \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

We will re-visit the inverse matrix later to study it properly, but here we just admit that B^{-1} of our matrix B exists with the property we want.

Definition 4. For a given $A \in \mathbb{R}^{3 \times 3}$, if there exists a matrix $L \in \mathbb{R}^{3 \times 3}$ such that

$$AL = LA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which must be then unique, L is said to be the inverse matrix of A , and denoted by A^{-1} .

1. Suppose that I have the Dictionary and suppose I also know its inverse. Then I can refer to any location in both language.
2. Then, if we get back to the reflection problem, we can do:
 - (i) Translate the location to his language.
 - (ii) Do the elementary reflection we just learned, that is along his axes.
 - (iii) Since the reflected result will be in his language, I translate it into my language.

That is to perform, for given $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$,

$$\begin{aligned}
 \begin{pmatrix} a \\ b \\ c \end{pmatrix} &\mapsto \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} && \text{Korean to English} \\
 \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} &\mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} = \begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} && \text{Elementary reflection in his language} \\
 \begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} &\mapsto \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} = \begin{pmatrix} d \\ e \\ f \end{pmatrix} && \text{English to Korean}
 \end{aligned}$$

This is written all at once,

$$\begin{pmatrix} d \\ e \\ f \end{pmatrix} = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

the reflected location in my language.

Chapter 6

Linear mappings from X to Y

- We will re-visit $\mathbb{R}^{n \times n}$ of square matrices, to discuss the determinant of a square matrix, the inverse if exists, how we can calculate them, and so on.
- Continuing discussions on the question: What does the matrix A do ?
- On the matrix $A \in \mathbb{R}^{m \times n}$ that is not square.
- A matrix in $\mathbb{R}^{n \times n}$ is called a square matrix. A matrix in $\mathbb{R}^{m \times n}$ is called a rectangular matrix. Of course, a square matrix is a rectangular matrix.

- Because we have developed the same on : “what does a square matrix A do?”,
- We can also easily understand “what does a rectangular matrix do?” too.
- This is done through the notion of the rank of matrix.

First of all, in this chapter,

- $\mathbb{R}^{n \times 1}$ represent a vector space X of dimension n ,
- $\mathbb{R}^{m \times 1}$ represent another vector space Y of dimension m ,
- $\mathbb{R}^{m \times n}$ represent the space of linear mappings from X to Y .

Rank of a matrix A

Let $A \in \mathbb{R}^{m \times n}$, i.e., it has m rows and n columns, in the way

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix}.$$

Motivation and observation: when the number “one” is relevant

Consider a special matrix $A \in \mathbb{R}^{3 \times 5}$,

$$A = \begin{pmatrix} 1 & 2 & 10 & 200 & 3 \\ 2 & 4 & 20 & 400 & 6 \\ 3 & 6 & 30 & 600 & 9 \end{pmatrix}$$

so that every columns are constant multiple of the first column.

But, it is also such that every rows are constant multiple of the first row.

Try differently. Is this always so ?

Motivation and observation: when the number “two” is relevant

Consider a special matrix $A \in \mathbb{R}^{2 \times 3}$,

$$A = \begin{pmatrix} 3 & 0 & -9 \\ 1 & 2 & 1 \end{pmatrix}$$

Certainly, $(3, 0, -9)$ and $(1, 2, 1)$ are the two independent rows.

Are the three columns independent ?

No. Certainly, $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ are linearly independent.

However, $\begin{pmatrix} -9 \\ 1 \end{pmatrix} = -3 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, i.e., $\begin{pmatrix} -9 \\ 1 \end{pmatrix} \in \text{span} \langle \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \rangle$.

Thus, the number of linearly independent columns coincide the number “two”.

It is not obvious to see why, but the number of linearly independent rows and the number of linearly independent columns seem to be the same.

To be able to check the statement in mathematical way, we do the followings.

1. We collect

$$\text{span} \langle n \text{ columns of } A \rangle \subset \mathbb{R}^{m \times 1}.$$

2. And we also collect

$$\text{span} \langle m \text{ rows of } A \rangle \subset \mathbb{R}^{1 \times n}.$$

The question is how many number of linearly independent elements in them ?

Before we proceed, we introduce two notations.

- For $A \in \mathbb{R}^{m \times n}$, A^T , the transpose of A , is the matrix in $\mathbb{R}^{n \times m}$ where the rows and columns are interchanged, i.e.,

$$(A^T)_{ij} = A_{ji}, \quad j = 1, 2, \dots, m, \quad i = 1, 2, \dots, n.$$

- For two columns $x, y \in \mathbb{R}^{\ell \times 1}$, we say x and y are orthogonal to each other if

$$x^T y = 0.$$

We now give a name for the above two we collected.

1. We call the first one, as

$$\text{Ran}(A), \quad \text{the range of } A$$

2. We consider the *transpose* the second one,

$$\text{Trow}(A) = \text{span} \langle m \text{ transpose of rows of } A \rangle \subset \mathbb{R}^{n \times 1}.$$

The range $\text{Ran}(A)$ is literally the range of A .

- Check: if $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, then according to the Matrix Multiplication rule,
 $Ax = x_1c_1 + x_2c_2 + \cdots x_nc_n$, a linear combination of columns of A .
- Here, $c_1, c_2, \dots, c_n$ are columns of A .

A subset $E \subset X$ is called a vector subspace of X if E is also a vector space.

$\text{Ran}(A) \subset \mathbb{R}^{m \times 1}$ is a *vector subspace* of $\mathbb{R}^{m \times 1}$, and $\text{Trow}(A) \in \mathbb{R}^{n \times 1}$ is a *vector subspace* of $\mathbb{R}^{n \times 1}$.

The objective of the next lecture is to reveal the following:

1. $\text{Trow}(A)$ and $\text{Ran}(A)$ are both r -dimensional, and
2. “The question we posed” can be understood by looking at the square matrix of $\mathbb{R}^{r \times r}$

$$\text{Trow}(A) \rightarrow \text{Ran}(A).$$

between the two.

3. What does the square matrix do? We completely know.

This is known as the Fundamental Theorem of Linear Algebra. We will go through this in the next lecture in detail.

Chapter 7

Fundamental Theorem of Linear Algebra

Here, we verify the picture:

through the fundamental theorem of linear algebra. We divide it into two parts.

We recall the orthogonality.

Theorem 1. Let $x_0 \in \mathbb{R}^{n \times 1}$.

$$x_0 \perp \text{Trow}(A) \iff Ax_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Proof.

$$\begin{aligned} x_0 \perp \text{Trow}(A) &\iff x_0^T (\lambda_1 b_1 + \lambda_2 b_2 + \cdots + \lambda_m b_m) = 0, \quad b_1, b_2, \dots, b_m \text{ are transpose of rows of } A \\ &\iff x_0^T b_i = 0 \quad \text{for every } i = 1, 2, \dots, m \\ &\iff i\text{-th component of } Ax_0 = 0 \text{ for every } i = 1, 2, \dots, m, \end{aligned}$$

according to the Matrix Multiplication Rule. □

Theorem 2. For any matrix $A \in \mathbb{R}^{m \times n}$,

the number of lin. indep. elements in $\text{Trow}(A)$ = the number of lin. indep. elements in $\text{Ran}(A)$.

- In fact we only need to check that for every matrix $A \in \mathbb{R}^{m \times n}$,
the number of lin. indep. elements in $\text{Trow}(A) \leq$ the number of lin. indep. elements in $\text{Ran}(A)$.
- This is because, if that is true, then do the same on another matrix $B = A^T$,
the number of lin. indep. elements in $\text{Trow}(A^T) \leq$ the number of lin. indep. elements in $\text{Ran}(A^T)$.

Proof. We only check the inequality:

1. To do that: Suppose $b_1, b_2, \dots, b_r \in \text{Trow}(A)$ are linearly independent. We show that $a_1 = Ab_1, a_2 = Ab_2, \dots, a_r = Ab_r$ are linearly independent.
2. Method by proving contrapositive statement.
3. Suppose that $a_1, a_2, \dots, a_r$ are linearly dependent. i.e., we can find coefficients $\lambda_1, \lambda_2, \dots, \lambda_r$, not all of them are zero, such that

$$\lambda_1 a_1 + \lambda_2 a_2 + \dots + \lambda_r a_r = 0.$$

4. In other words, by linearity,

$$A(\lambda_1 b_1 + \lambda_2 b_2 + \dots + \lambda_r b_r) = 0.$$

5. But Theorem 1 says that then

$$x_0 = \lambda_1 b_1 + \lambda_2 b_2 + \dots + \lambda_r b_r \in \text{Trow}(A) \text{ is orthogonal to all of } \text{Trow}(A).$$

6. The only element in $\text{Trow}(A)$ that is orthogonal to all of $\text{Trow}(A)$ is only

$$x_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

7. This concludes that $b_1, b_2, \dots, b_r \in \text{Trow}(A)$ are linearly dependent.

□

Chapter 8

Computing by Hands: The Picture of Fundamental Theorem of Linear Algebra

In this chapter, we discuss about following questions. Suppose a matrix $A \in \mathbb{R}^{m \times n}$ is given.

1. How to pick up basis elements of $\text{Ran}(A)$?
2. How to pick up basis elements of $\text{Trow}(A)$?
3. How to pick up basis elements of $\text{Trow}(A)^\perp$?
4. How to pick up basis elements of $\text{Ran}(A)^\perp$?
5. How to calculate the rank r ?

In case the shape of the matrix A is of very big size, in general, those questions are subjected to running some algorithm using computer.

However, if the size of the matrix A is not too large, you would want to do it by your hands with pen and paper. Today, we introduce one technique starting with two Good News.

1. In fact, you already know this.
2. The way it achieves is so nice that you would extract good enough information from a certain amount of work.

Row operation: Replacement of A by A'

- Here, we consider A' of same shape, instead of A , anyone satisfying $\text{Trow}(A') = \text{Trow}(A)$, i.e., the **span** of rows of A' and the **span** of rows of A are the same.
- You are free to change the order of rows.
- One typical example is to pick up one of the row of A , and replace it by the row itself + a linear combination of remaining rows of A .
- We could equivalently do the Column operation. Traditionally, this technique has been done in the way of Row Operation.

Here comes the first good news, that you already know this:

Example

$$\begin{cases} 2x - 3y = 5 \\ x - 2y = 2 \end{cases} \quad \text{Let } A = \begin{pmatrix} 2 & -3 & 5 \\ 1 & -2 & 2 \end{pmatrix}$$

1. According to the Matrix Multiplication Rule, Row operation is implemented by multiplying $m \times m$ matrix from the left.
2. For our purposes, it is matter of choice whether you involves those $m \times m$ matrices or you omit them. Do what are convenient for you.

The purpose of the replacement, and why it is easy

1. We are here, replacing the matrix A sequentially by

$$A \rightarrow A' \rightarrow A'' \rightarrow A''' \rightarrow \dots$$

while keeping $\text{Trow}(A) = \text{Trow}(A') = \text{Trow}(A'') = \text{Trow}(A''') = \dots$.

2. Of course, those matrices appear later have to be simpler than ones appeared before, in what so ever senses. For example, there are more 0 in the entries. Or we can easily extract some basis elements of $\text{Trow}(A)$ out of it.
3. By contrast, the information on $\text{Ran}(A)$ are completely forgotten. In general,

$$\text{Ran}(A) \neq \text{Ran}(A'), \quad \text{Ran}(A') \neq \text{Ran}(A''), \quad \dots$$

and in this stage we simply do not care whatever happens in range.

$\Rightarrow$ The consequence then will be:

- We will see much simpler picture of $\text{Trow}(A)$,
- We will still keep the same $\text{rank}(A)$, thanks to the Fundamental Theorem.
- We will completely lose the picture of $\text{Ran}(A)$.
- But once the Row operation is finished, column picture will be so nicely recovered.
This is the second good news.
- We do this one by one.
- The goal of this is known as the Column-Row (CR)-decomposition, that can be seen in the Lecture of G. Strang.

What do we do ?:

1. Introduce 0 as many as possible.
2. Turn it to the row echelon form.
3. Turn it to the reduced row echelon form.
4. Check if calculations went correct: Complete the Column-Row decomposition.

Example 2

$$A = \begin{pmatrix} 1 & 3 & 10 & 200 \\ 2 & 6 & 20 & 400 \\ 3 & 9 & 30 & 600 \end{pmatrix}$$

Example 3

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 7 \\ 1 & 4 & 6 \end{pmatrix}$$

Example 4

$$A = \begin{pmatrix} 0 & -7 & -4 & 2 \\ 2 & 4 & 6 & 12 \\ 3 & 1 & -1 & -2 \end{pmatrix}$$

Example 5

$$A = \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix}$$

More on computation by hands

Example: Complete the CR decomposition of

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 7 \\ 1 & 4 & 6 \end{pmatrix}$$

Summarize: For example, we arrived at

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 7 \\ 1 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

1. $\text{rank}(A)$: (RHS) is a product of (3×2) and (2×3) matrices. $\text{rank}(A) = 2$.
2. $\text{Trow}(A)$: $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$ is in much simpler form. Big improvement.
3. $\text{Ran}(A)$: $\begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 1 & 4 \end{pmatrix}$ at least reveals which columns are independent. Small improvement

1. We already know the $\text{rank}(A)$.
2. We select basis elements of the domain $\mathbb{R}^{n \times 1}$, adapted to $\text{Trow}(A)$.
3. We select basis elements of the codomain $\mathbb{R}^{m \times 1}$, adapted to $\text{Ran}(A)$.

Selecting basis elements of the domain $\mathbb{R}^{n \times 1}$, adapted to $\text{Trow}(A)$

1. We select basis of $\text{Trow}(A)$.

Out of the last form

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

We select two elements

$$\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

that span $\text{Trow}(A)$.

2. Now, we select basis of $\text{Trow}(A)^\perp$.

- (a) This is to find element $x_0 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ such that

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

- (b) This equation is super easy to solve.

- (c) Recalling that the 1st and 2nd columns are marked independent columns, we put $c = -1$, and let $\begin{pmatrix} a \\ b \end{pmatrix}$ be the precisely $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$. This gives rise to

$$\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

We have selected all the basis for the domain:

$$\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

The first two span $\text{Trow}(A)$, and the last span $\text{Trow}(A)^\perp$. They are not orthogonal to each other though.

Selecting basis elements of the codomain $\mathbb{R}^{m \times 1}$, adapted to $\text{Ran}(A)$

- Swapping the role of row and column, I consider A^T , and turn A^T into the Reduce Row Echelon Form.
- In fact, it suffices to turn C^T into the Reduced Row Echelon Form.
- Let us do that. Out of C^T ,

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \end{pmatrix}$$

- We turn it into

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \end{pmatrix}$$

1. Now, the we select basis of $\text{Ran}(A)$.

$$\begin{pmatrix} 1 \\ 0 \\ -5 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}.$$

2. Select basis of $\text{Ran}(A)^\perp$:

$$\begin{pmatrix} -5 \\ 3 \\ -1 \end{pmatrix}.$$

We have selected all the basis for the domain:

$$\begin{pmatrix} 1 \\ 0 \\ -5 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} -5 \\ 3 \\ -1 \end{pmatrix}.$$

The first two span $\text{Ran}(A)$, and the last span $\text{Ran}(A)^\perp$. They are not orthogonal to each other though.

If you also have completed the decomposition of C^T ,

$$C^T = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \end{pmatrix},$$

You may write

$$A = CR = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

Home Work

Do each of following steps for the matrices written below.

1. Turn the matrix A into the Reduced Row Echelon Form.
2. Complete the $A = CR$ decomposition, checking if the calculation was correct.
3. Turn the matrix C^T into the Reduced Row Echelon Form.
4. Compute the rank of A .
5. Select basis elements of domain $\mathbb{R}^{n \times 1}$:
 - (a) basis elements of $\text{Trow}(A)$,
 - (b) basis elements of $\text{Trow}(A)^\perp$.
6. Select basis elements of codomain $\mathbb{R}^{m \times 1}$:
 - (a) basis elements of $\text{Ran}(A)$,
 - (b) basis elements of $\text{Ran}(A)^\perp$.

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 5 \\ 2 & 4 & 6 & 9 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Chapter 9

System of m equations and n unknowns

We consider a system of linear equations:

$$\begin{aligned}A_{1,1}x_1 + A_{1,2}x_2 + A_{1,3}x_3 + \cdots + A_{1,n}x_n &= b_1 \\A_{2,1}x_1 + A_{2,2}x_2 + A_{2,3}x_3 + \cdots + A_{2,n}x_n &= b_2 \\&\vdots \\A_{m,1}x_1 + A_{m,2}x_2 + A_{m,3}x_3 + \cdots + A_{m,n}x_n &= b_m\end{aligned}$$

We write the system

$$Ax = b,$$

where

$$A \in \mathbb{R}^{m \times n}, \quad x \in \mathbb{R}^{n \times 1} = X, \quad b \in \mathbb{R}^{m \times 1} = Y.$$

One might think that a meaningful solution map

$$R : Y \rightarrow X, \quad b \mapsto x = Rb$$

is only possible when $m = n$ and actually $m = n = r$, with $r = \text{rank}(A)$.

However, this is not true. You know everything to give a complete set of the least square solutions for any m , n , and the rank r .

Two Obvious Facts

1. If $b \notin \text{Ran}(A)$, then there can be no way to have any solution.
2. Suppose $b \in \text{Ran}(A)$.
 If $\text{Trow}(A) \neq \mathbb{R}^{n \times 1}$, i.e., $\text{Ker}(A) = \text{Trow}(A)^\perp$ is not trivial, then there are infinitely many solutions.
 This is because, that $b \in \text{Ran}(A)$ implies there exists $x \in \mathbb{R}^{n \times 1}$ with $Ax = b$.
 Now, for infinitely many choices of $x_0 \in \text{Ker}(A)$, $x + x_0$ is again a solution.

What do we want?

1. We make a choice, in Y , a projection onto $\text{Ran}(A)$ for any $b \in Y$.
2. Not surprisingly, $b \mapsto b' \in \text{Ran}(A)$ that is by the orthogonal projection is the one used mostly.
3. Now, for given the projection b' , we completely know that:
 - (a) There exists unique element $x' \in \text{Trow}(A)$ such that $Ax' = b'$.
 - (b) The complete set of *solutions* are $x' + x_0$ for any $x_0 \in \text{Ker}(A)$.
4. Any least square solution is the one where

$$Ax = b', \quad b' \text{ is the element in } \text{Ran}(A) \text{ such that } \|b - b'\|_2 \text{ is minimized.}$$

Solutions of $Ax = 0$, by RREF

Let

$$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$$

- We already know how to obtain basis of $\text{Trow}(A)$ and $\text{Ker}(A) = \text{Trow}(A)^\perp$.
- Let us repeat the procedure in detail again though.

Suppose we make a row operation into

$$A' = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 3 & 6 & 8 & 10 \end{pmatrix}$$

This is to do

$$EA = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 3 & 6 & 8 & 10 \end{pmatrix} = A'$$

and we recall that

1. We multiply E such that

$$\text{Trow}(A) = \text{Trow}(EA), \quad \text{and consequently} \quad \text{Trow}(A)^\perp = \text{Trow}(EA)^\perp.$$

2. Hence, the complete set of

$$\{x \in \mathbb{R}^{n \times 1} \mid Ax = 0\} = \text{Trow}(A)^\perp = \text{Trow}(EA)^\perp = \dots = \{x \in \mathbb{R}^{n \times 1} \mid Ux = 0\},$$

where U is the RREF of A .

Chapter 10

Solutions of $Ax = b'$ when $b' \in \text{Ran}(A)$

- In this chapter, we calculate solutions of the equation, when we do know

$$b' \in \text{Ran}(A).$$

We keep working with

$$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$$

and take

$$b' = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

to solve $Ax = b'$.

- If x solves the equation $EAx = Eb'$, and E is a row operation, then x solves

$$Ax = b',$$

because E is always invertible.

- Therefore, in searching for any solution x , the same operation by E has to be applied on the (r-h-s) too, keeping

$$(A, b) \rightarrow (E_1 A, E_1 b') \rightarrow (E_2 E_1 A, E_2 E_1 b') \rightarrow \dots$$

- Or, we append the b' column to A and multiply row operation E to the whole:

$$\left(\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 2 & 4 & 6 & 8 & 1 \\ 3 & 6 & 8 & 10 & 2 \end{array} \right)$$

- In preceding chapter, we considered $Ax = 0$, and there we might have also appended 0 column, which has no effect.

REF or RREF of the augmented matrix

1. Let us first turn the matrix into REF:

$$\left(\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

2. Observations:

- (a) We observe the 1st and 3rd columns are the pivot column, and in writing

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix},$$

we called the x_1 and x_3 the primary or pivot variables, and the remaining x_2 and x_4 the free variables.

- (b) Also, we see the last row is 0 row, including the (r-h-s) part. Otherwise, $b' \notin \text{ran}(A)$.

3. Backsubstitution.

- (a) We first put all free variables 0.

$$\text{We look for a particular solution } x_p = \begin{pmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{pmatrix}.$$

(b) If so, we have

$$2x_3 = -1 \iff x_3 = -\frac{1}{2}$$

(c) Now,

$$x_1 + 2x_3 = 1, \quad x_3 = -\frac{1}{2} \implies x_1 = 2.$$

$$\text{A particular solution } x_p = \begin{pmatrix} 2 \\ 0 \\ -\frac{1}{2} \\ 0 \end{pmatrix}.$$

4. We recall that

$$\text{Ker}(A) = \text{span} \left\langle \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\rangle,$$

in other words

$$c_1 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

span all solutions of $Ax = 0$.

5. Thus,

$$x = \begin{pmatrix} 2 \\ 0 \\ -\frac{1}{2} \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

comprises the complete set of solutions for $Ax = b'$.

6. The particular solution we found is not a member of $\text{Trow}(A) = \text{Ker}(A)^\perp$ though.

1. Suppose we further turn REF to RREF,

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -2 & 2 \\ 0 & 0 & 1 & 2 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

2. Reading off a particular solution

- (a) We first put all free variables 0.

We look for a particular solution $x_p = \begin{pmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{pmatrix}$.

- (b) If so,

A particular solution $x_p = \begin{pmatrix} 2 \\ 0 \\ -\frac{1}{2} \\ 0 \end{pmatrix}$.

Algorithm summary

- Turning $(A|b')$ into REF, and solving by back substitution can be put into the algorithm:

known as the row-pivoting (row swap allowed) Gauss elimination.

- In the algorithm, we turn

$$E_n E_{n-1} \cdots E_1 A = U,$$

where each of E_j is either row swap or row replacement.

- Once REF U is obtained, we do the backsubstitution.
- Backsubstituting a rectangular REF U means, we suitably treat the pivot and free variables to solve

$$Ax_0 = 0, \quad \text{and} \quad Ax_p = b'.$$

- Turning $(A|b')$ into RREF is known as the row-pivoting (row swap allowed) Gauss-Jordan elimination.

Chapter 11

Orthogonal Projection of b onto $b' \in \text{Ran}(A)$

- Now, suppose we are given $b \in \mathbb{R}^{m \times 1}$, but we do not know if $b \in \text{Ran}(A)$ or not.
- Here, we look for $b' \in \text{Ran}(A) \subset \mathbb{R}^{m \times 1}$ that is the orthogonal projection of b onto $\text{Ran}(A)$.
- Once b' is obtained, we solve complete set of solutions for $Ax = b'$ by turning $(A \mid b')$ into RREF.

- This leads to solving the least square problem
 1. Of course, the distance $|b - b'|$ is minimum distance among all distances $|b - c|$, $c \in \text{Ran}(A)$.
 2. The Euclidean length

$$|b - b'| = \sqrt{(b_1 - b'_1)^2 + \cdots + (b_m - b'_m)^2}.$$

Orthogonal Projection on $\text{Ran}(A)$

- To compose the orthogonal projection, what we need is the basis $w_1, w_2, \dots, w_r$ that span $\text{Ran}(A)$ and basis $w_{r+1}, w_{r+2}, \dots, w_m$ that span $\text{Ran}(A)^\perp$.
- We at least know one way to calculate $w_1, w_2, \dots, w_r$, and also $w_{r+1}, w_{r+2}, \dots, w_m$, using RREF.
- We also know how to compose the projection.
- So, we at least know one way to compute b' .

- In other words: we are given

1. Linearly independent $\mathbb{R}^{m \times 1}$ columns $w_1, w_2, \dots, w_m$
2. such that

$$\text{span} \langle w_1, w_2, \dots, w_r \rangle = \text{Ran}(A), \quad \text{span} \langle w_{r+1}, w_{r+2}, \dots, w_m \rangle = \text{Ran}(A)^\perp,$$

(By definition of $\text{Ran}(A)^\perp$, $\text{span} \langle w_1, w_2, \dots, w_r \rangle \perp \text{span} \langle w_{r+1}, w_{r+2}, \dots, w_m \rangle$).

- If $w_1, w_2, \dots, w_m$ are so,

1. Let (English-To-Korean Dictionary)

$$W = \begin{pmatrix} | & | & \cdots & | \\ w_1 & w_2 & \cdots & w_m \\ | & | & \cdots & | \end{pmatrix}$$

2. Let the $m \times m$ elementary projection

$$\hat{P} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

3. The projection

$$P = W\hat{P}W^{-1}$$

will give $b \mapsto b' = Pb$.

A few comments before entering calculation

1. We will do this today using what we have learned so far.
2. While to obtain the basis out of RREF, and CR decomposition of A is good because we can achieve this by hand,
3. CR decomposition is usually for pedagogical purposes, not for real problems.
4. Motivated from this, we will introduce QR decomposition and SVD for the purpose of obtaining (orthonormal) basis, for real world problems.

5. Another remark is the following: One might have seen, in solving the least square problem, the system

$$A^T Ax = A^T b.$$

- (a) As a matter of fact, this works in the special case where the matrix is thin and full rank:

$$m \geq n = r.$$

- (b) In that case, we will clarify how this is a projection-involved least square problem.

We work again with

$$\begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$$

with

$$b = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

This consists of six steps:

1. Choose basis for $\text{Ran}(A)$.
2. Choose basis for $\text{Ran}(A)^\perp$.
3. Compute the projection matrix P and compute $b' = Pb$.
4. Do the RREF for the augmented matrix.
5. Compute all the solutions $Ax = 0$.
6. Compute a particular solution of $Ax_p = b'$

Home Work: Obtain all the least square solutions for

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 10 \\ 3 & 6 & 9 & 12 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}.$$